

Network automation graveyard

Field-tested anti-patterns to kill your project

AUTOCON S

THE NETWORK AUTOMATION CONFERENCE

Introduction

About me

- Constantin VURLI GUILABERT
- Net. Automation Consultant at CNS Communications
- 8 years in networking
- 5 years focused on automation

About this talk

Focus

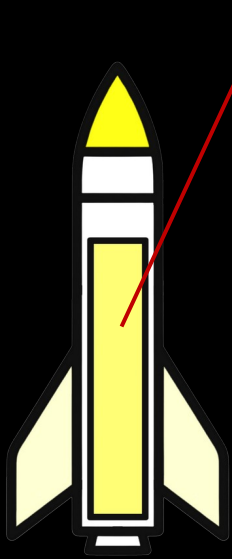
- Recurrent patterns
- Actionable situations

Biases

- European market
- Medium to large orgs
- Enterprise mostly

You've seen the moon landings. Here are the crashes.

The Network Automation Rocket



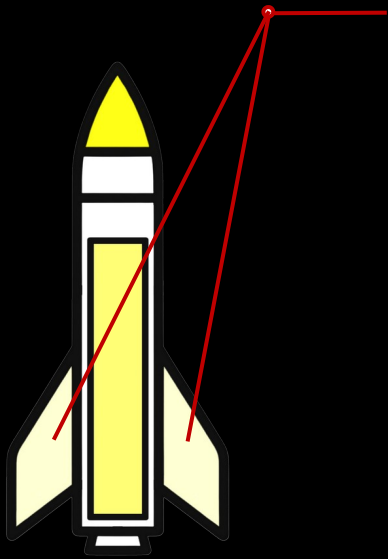
Fuel

- Budget & Investment
- Time/headcount allocated
- Sponsoring for decisions

Fuel is expensive :

more fuel = more expectations

The Network Automation Rocket

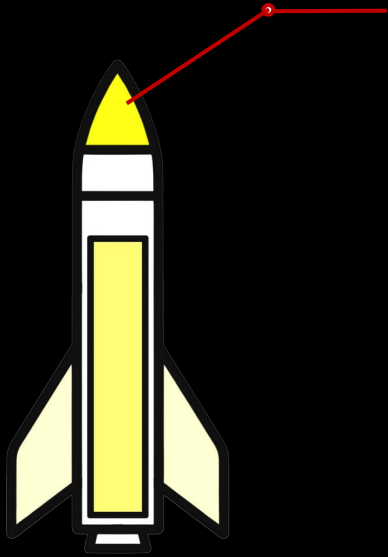


Thrust

- Skills level & diversity
- Time spent efficiently
- The right tooling

Having fuel does not mean
you are burning it

The Network Automation Rocket

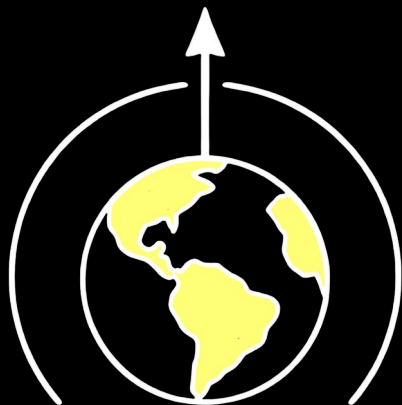


Guidance system

- Architecture decisions and technical roadmap
- Use case prioritization
- Trajectory correction

Full thrust in the wrong direction still misses orbit

The Network Automation Rocket



Mass

- Complexity of your project

Gravity

- Resistance of your org

You need to reach escape
velocity

The Graveyard

The ones that didn't make it

The Pontiac fiero



The Pontiac fiero



Flight report

Fuel:KO|Thrust:OK|Guidance:OK

Cause : Under-engineering the platform

Symptoms :

- Break production
- Spend 90% of time fixing stuff
- Stuck when you want to go to

Prognosis : Exploded in flight

The Pontiac fiero



Flight report

Fuel:KO|Thrust:OK|Guidance:OK

Cause : Under-engineering the platform

Symptoms :

- Break production
- Spend 90% of time fixing stuff
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Prognosis : Exploded in flight

The Pontiac fiero : Victim report

How did you end up here ?

- Automation initiative starts on top of existing tooling
- Each new use case requires a workaround on top of a workaround
- Velocity increases until you hit the structural limits, then it explodes mid-flight

Typical victim

- Small/Highly technical teams with lots of autonomy and no sponsoring from top management
- No greenfield opportunity, no budget for re-architecture

The Pontiac fiero : How to fix

Short term fix :

- Get management to understand the risks/reward
- Use the right language & metrics to get : Long term investment & Sponsorship

Long term fix :

- Use the NAF Framework & an opportunistic trajectory to refactor
- Identify and present good « side effects » of automation that sweeten the investment

The Imperial Star Destroyer



The Imperial Star Destroyer



Flight report

Fuel:OK|Thrust:KO|Guidance:OK

Cause : Over-engineering the platform

Symptoms :

- Missing deadlines
- Low business value

Prognosis : Project cancelled

The Imperial star destroyer



Flight report

Fuel:OK|Thrust:K0|Guidance:OK

Cause : Over-engineering the platform

Symptoms :

- Missing deadlines
- Low business value

Prognosis : Project cancelled

The ISD : Victim report

How did you end up here ?

- Early hard problems feel productive (HA, multi-tenancy, API-first everything)
- Each new requirement adds a layer, nobody says stop 18 months in, the platform is impressive and the business has automated nothing

Typical victim

- Network engineering team that finally got budget and went all-in
- Nobody dares to say « *this is too much* »

The ISD : How to fix

Short term fix :

- Find a use case, and just do it
- Find another one, and just do it
- Report results

Long term fix :

- Use the NAF Framework (yes, again)
- Iterate on the rocket (first go to the moon, then to)
- Always ask, « do I really need that TODAY »

The Shuttle



The Shuttle



Flight report

Fuel:OK|Thrust:OK|Guidance:KO

Cause : Wrong flight plan

Symptoms :

- Still many manual operations
- Low level of abstraction
- No toil reduction

Prognosis : Did 30 launches, no one on the moon yet

The Shuttle



Flight report

Fuel:OK|Thrust:OK|Guidance:KO

Cause : Wrong flight plan

Symptoms :

- Still many manual operations
- Low level of abstraction
- No toil reduction

Prognosis : Did 30 launches, no
one on the moon yet

The Shuttle : Victim report

How did you end up here ?

- First automation wins are celebrated
- Business asks for more of the same, team delivers more of the same
- Comfort zone becomes the ceiling

Typical victim

- Team under constant pressure to deliver, zero slack for investment.
- The team knows it's not enough, but nobody gives them permission to think bigger.

The Shuttle : How to fix

Short term fix :

- Measure toil, not just tickets closed
- Show what's *not* being automated and what it costs
- Find one use case that's above your ceiling and do it

Long term fix :

- Get the mandate, not just the budget
- Reframe the team's identity: ops → platform

Victims of gravity



Too many rockets
(I'll do it on my own)



The eternal drawing board
(Should we make a ninth meeting ?)

Get help from a consulting firm (for real)

The right rocket

How to find balance ?

A balancing act

Complexity

- Not complex enough, you will not reach the objectives
- Too complex, you will stall
- Simplicity is a hard feature to design & implement

Time horizon

- Think short term, you'll end up redesigning from scratch
- Think long term, you'll be disconnected from reality

The right rockets

Iterate

- Vertical : 1 use case end to end, small perimeter
- Horizontal : 1 feature, all perimeter

Look at other rockets

- Join a community, get feedback
- But stay focused on your context

Perfection is achieved not when there is nothing more to add,
but when there is nothing left to take away.

Antoine de Saint-Exupéry



Questions?